

Watin Promfiy

DATA SCIENCE / AI ENGINEER / DATA ENGINEER

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PROFESSIONAL SUMMARY

Data science student at KMITL (GPA 3.87) and in Musashino University's integrated undergraduate–master's program, with industry experience in machine learning, LLMs, and predictive maintenance at Asia Energy Tech, Bot and Life, and KBTG, and teaching roles at AIAT, KMITL, and DGA. First author of five international conference papers accepted in 2026 and a three-time hackathon winner, including the BDI Hackathon Royal Cup.

EDUCATION

- Integrated Undergraduate–Master's Program in Data Science** Sep 2026 – Present
Musashino University, Japan · Faculty of International Data Science and Graduate School of Data Science
- B.Sc. in Information Technology (Data Science and Business Analytics)** 2023 – Present
King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand · Cumulative GPA: 3.87/4.00

PROFESSIONAL EXPERIENCE

- AI Engineer Intern**, Asia Energy Tech (AET) Jul 2026 – Present
- Developed predictive maintenance models for industrial pump sensor data, benchmarking 13 detection methods.
 - Built an equipment health dashboard and an LLM maintenance-report generator with automated fact verification.
- AI Engineer (Part-Time)**, Bot and Life Jul 2025 – Feb 2026
- Engineered a multi-agent platform that automates product banner and marketing asset creation.
 - Implemented prompt engineering strategies that expand user inputs in context to improve generated content.
- LLM & Prompt Engineer Intern**, Bot and Life Nov 2024 – Feb 2025
- Architected an LLM-powered task management system with memory for long-term conversational context.
 - Developed prompt frameworks to minimize hallucination and enforce structured output compliance.
- Data Science Intern (Apprentice Program)**, KBTG (K-LAB) Jul – Dec 2024
- Developed a PyTorch model estimating roof surface area and solar energy potential from satellite imagery.
 - Engineered a condominium similarity engine using geospatial data and deep learning for property valuation.

TEACHING EXPERIENCE

- Teaching Assistant**, King Mongkut's Institute of Technology Ladkrabang (KMITL) Jun – Sep 2026
Applied Machine Learning and Data Warehouse courses (DSBA8)
- Developed weekly labs on scikit-learn machine learning and Airflow, dbt, and PostgreSQL data warehouse pipelines.
 - Built a student progress dashboard for lab grades and a synthetic retail dataset for a CRISP-DM assignment.
- Teaching Assistant**, Digital Government Development Agency (DGA) May – Aug 2026
e-Government Executive Program, Cohorts 1–3
- Mentored participants in analyzing Thai open government data with Python, Plotly, and machine learning models.
 - Compiled a public data visualization catalog of the interactive dashboards produced across three cohorts.
- Teaching Assistant**, Artificial Intelligence Association of Thailand (AIAT) Apr – Jul 2025; May – Jul 2026
Super AI Engineer Program, Seasons 5 and 6
- Delivered curriculum from machine learning to LLMs, including LLM inference and LoRA/QLoRA sessions on LANTA.
 - Mentored participants in Python debugging and led paper-reading sessions on state-of-the-art AI research.

TECHNICAL SKILLS

AI & LLMs: LLM fine-tuning (LoRA/QLoRA), RAG, VLMs, MoE, multi-agent systems, prompt engineering, MLX
Machine Learning: Transformers, CNNs, LSTMs, gradient boosting, anomaly detection, time-series forecasting
Data Science: Statistical modeling, feature engineering, geospatial analysis, knowledge graphs, Plotly
Data Engineering: ETL/ELT, PySpark, Airflow, Kafka, dbt, PostgreSQL, BigQuery, Snowflake
MLOps & Cloud: FastAPI, Docker, inference optimization, vector databases, HPC, AWS, GCP, Huawei Cloud
Programming: Python, SQL, TypeScript, Git, Linux
Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, LlamaIndex, Next.js, n8n

PUBLICATIONS

Publication Summary: 6 accepted international conference papers (5 as first author), 3 conference papers under review, and 2 journal manuscripts in preparation.

Peer-Reviewed International Conference Papers (Accepted, 2026)

- [1] **W. Promfiy**, P. Boonprasart, V. Sornlertlamvanich, and S. Poomrittigul, "HIKARI: A Multi-Stage Vision-Language Prototype with RAG-in-Training for Medical Lesion Diagnosis," in *Proc. International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC)*, 2026.
- [2] **W. Promfiy**, T. Jaempinit, S. Koseeyaumporn, and S. Tarnpradab, "Yeeping-Eco: A Knowledge Graph RAG Framework with Predictive PM2.5 Routing," in *Proc. International Conference on Big Data Analytics and Practices (IBDAP)*, 2026.
- [3] S. Koseeyaumporn, **W. Promfiy**, S. Saiprom, and S. Tarnpradab, "MANJU: A Multi-Agent Framework for Natural Just-in-Time Understanding in Thai," in *Proc. International Joint Conference on Computer Science and Software Engineering (JCSSE)*, 2026.
- [4] **W. Promfiy** and S. Poomrittigul, "Agents Optional, Calibration Essential: Efficient Medical Question Answering with Open Language Models," in *Proc. Biomedical Engineering International Conference (BMEiCON)*, 2026.
- [5] **W. Promfiy**, S. Poomrittigul, and W. Buavirat, "Deep-Unfolded Compressed Sensing for Energy-Efficient Wearable ECG: Diagnostic Preservation, Learned Binary Sensing, and Cross-Dataset Generalization," in *Proc. International Conference on Information Technology and Electrical Engineering (ICITEE)*, 2026.
- [6] **W. Promfiy** and A. Kuiprasert, "ZoonoMoE: A Voice-Driven Zoonotic Disease Surveillance System via Mixture-of-Experts Routing and Domain-Partitioned Retrieval-Augmented Generation," in *Proc. International Conference on Neural Information Processing (ICONIP)*, 2026.

International Conference Papers Under Review

- [7] **W. Promfiy**, S. Jitraveewon, and P. Thusaranon, "Mahatsachan or Mirage? Null-Matched Auditing of Minimal Affective Prompt Stimuli in Multilingual LLMs," submitted to *International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP 2026)*.
- [8] **W. Promfiy**, C. Sirikayon, P. Tuwanut, and S. Poomrittigul, "PREFER: Persistent-Frequency KV-Cache Eviction for LLM Prefix Caching," submitted to *International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP 2026)*.
- [9] **W. Promfiy**, A. Kuiprasert, and L. Pradittasnee, "Quantization Calibration and Speculative Decoding for On-Device Thai LLMs: A Measurement Study," submitted to *International Conference on Information Technology (InCIT 2026)*.

Journal Manuscripts in Preparation

- [10] **W. Promfiy** (first author), "Calibrate in the State You Deploy: State-Matched Post-Training Quantization for LLMs with Compressed KV Caches," manuscript in preparation.
- [11] S. Poomrittigul, D. Rungraungrayabkul, R. Unnaha, W. Sarakhon, **W. Promfiy**, B. Thanathornwong, S. P. Khovidhunkit, and T. Treebupachatsakul, "An Expert System for Oral Potentially Malignant Disorder Screening That Fuses Photographic Image Classification with Natural Language Processing of the Medical History," manuscript in preparation.
Contribution: Writing – original draft.

HONORS & AWARDS

Winner (Royal Cup) , BDI Hackathon – AI and Data Innovation for Smart Tourism (BDI, CMU) Knowledge-graph recommendation engine that generates personalized Chiang Mai itineraries.	2025
Winner , ExxonMobil Bootcathon (ExxonMobil) Agentic n8n workflow with text-to-Pandas natural-language data exploration.	2025
Winner , Krungsri UniVerse × KMITL Hackathon – Mule Hunt Challenge (Krungsri, KMITL) Mule-account network tracing with web crawling, graph visualization, and an LLM query interface.	2025
2nd Runner-Up , AI Thailand Hackathon (NECTEC) DAMZ: zero-shot object and action detection from Thai and English natural-language commands.	2025
2nd Runner-Up , Huawei ICT Competition – Cloud Track (Huawei Cloud) Top-three national placement in cloud deployment and AI/LLM examinations.	2025
Silver Medal , Super AI Engineer Season 4 (AIAT) Computer vision and predictive modeling solutions, including solar-potential and real estate projects at KBTG.	2024
1st Runner-Up , Krungsri UniVerse × KMITL Hackathon (Krungsri, KMITL) AI for Autism: personalized learning content generation for children with autism.	2024

Honorable Mention, AI Thailand Hackathon (NECTEC)	2024
Thai CV Analyzer for resume evaluation and job matching through reusable APIs.	
Top 15, JUMP THAILAND	2024
Transformer-based image captioning for visually impaired users.	
1st Runner-Up, Innovation for Crime Combating (I4C)	2023
Deepfake video detection pipeline using a CNN ensemble for digital forensics.	

SELECTED PROJECTS

RegMap AI – Regulatory Intelligence Platform	UN ESCAP RDTII · 2026
Mapped ASEAN digital-trade and data-protection laws to RDTII indicators using multilingual OCR and Llama-3.3-70B.	
KhonKaen Health × City Intelligence Platform	BDI Hackathon · 2026
Unified clinical, phenomics, and smart-city data in a knowledge graph with a cross-domain AI agent.	
Proactive Welfare Link (PWL)	Hack for Right · 2026
Automated welfare-tier assignment from ICD-10 diagnoses and Barthel ADL scores with PDPA-compliant tokenization.	
Nemotron-Personas-Thailand	NVIDIA × CMKL · 2026
Generated 300,000 synthetic Thai population records aligned with official statistics using NVIDIA NeMo Data Designer.	
JaiVoice – On-Device Thai Voice Assistant	2026
Built a real-time, interruptible Thai voice assistant running fully on a MacBook (MLX Whisper, Qwen, cloned-voice TTS).	
DiscordFlow – ML Experiment Logger	Python library · 2026
Developed a Python library that asynchronously logs training metrics, parameters, and artifacts to Discord.	
Agentic AI CV Generator	2026
Built a web app that turns text into LaTeX CVs with Qwen3-32B, a validation agent, and job matching.	
Skin Disease Image Scraper	2026
Built a scraper that collects up to 8,900 images across 178 skin diseases for medical image classification.	
Multi-Modal Deep Learning for Thai Fossil Classification	KMITL · 2025
Developed a multi-modal deep learning pipeline for fossil classification and paleontological analysis in Thailand.	
Home Credit – Credit Risk Model Stability	KBTG × AIAT Competition
Built gradient-boosting default-risk models with Gemini-assisted analysis, focused on stability over time.	
Table-Based Thai Question Answering	Super AI Engineer
Built question answering over tables using retrieval-augmented generation and LLM fine-tuning.	
Other projects: MacBook Vibration Player (Apple Silicon accelerometer), Thai TTS Discord bot, IoT sensor REST API (Hono, PostgreSQL), and OCR-based file organizer (Naive Bayes).	

ACADEMIC SERVICE

External Reviewer (Subreviewer): International Conference on Neural Information Processing (ICONIP 2026)